OmniCut personalized cutlery

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Project Diary and Analysis of Recontextualization

Summary of Weekly Diary

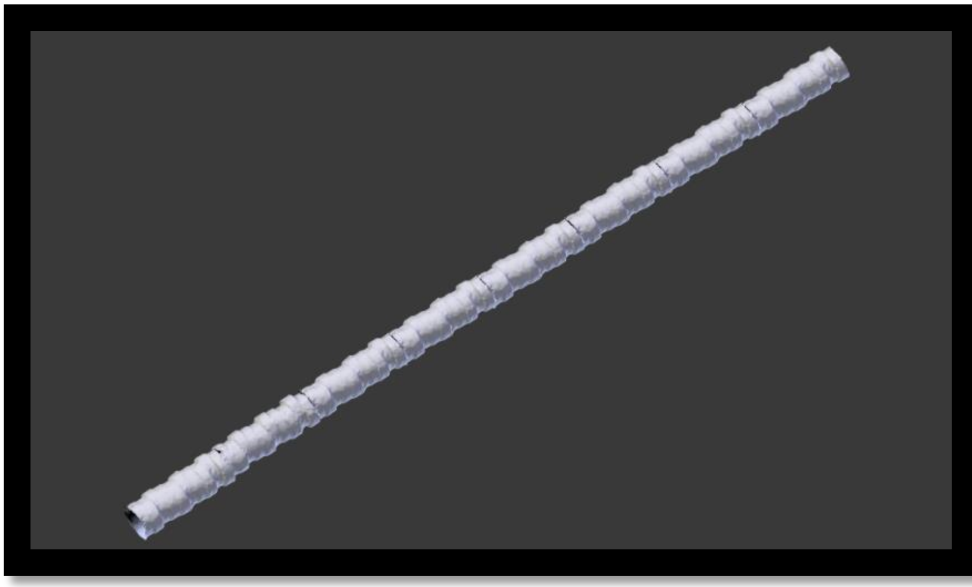
	Major Updates	Physical Computing
Week 1	-Chose object to recontextualize -Scanned lamp into computer -Momentous amounts of brainstorming happened -Decided on a set idea to pursue: set of chopsticks with cutlery attachments	-My partners looked around Baker Furniture and helped me scan the lamp I had chosen -At this point, the only ideas were possible lighting setups for display and other light play possibilities.
Week 2	-Refined the shape and taper of the chopsticks -Came up with a few methods for connection between attachment and base handle	-The primary idea is hatched: release small amounts of medicine into your food directly through the utensil.
Week 3	-Began modeling spoon attachment -Began modeling knife attachment -Decided on a more specific design element spanning the objects in the set	-The directions shifts towards different flavors instead of different medicines being dispensed
Week 4	-Continued and then finished the knife model -Began modeling the fork attachment -designed a wooden box to be cut using laser cutter	-Mostly ideas and deciding how exactly to achieve the goal happened
Week 5	-Finished up the models -Took care to resize the components correctly (but didn't really succeed)	-The pipette system was created and the motor was finally added to the electronic system

Week 1

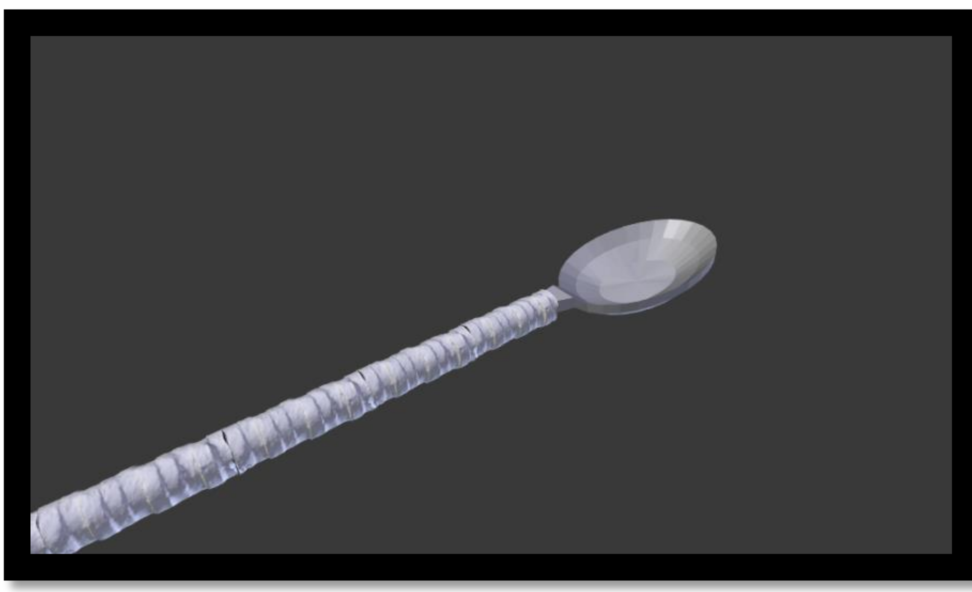
We first traveled to Boston Design Center to explore the area. We went through the participating stores to browse for a few hours. We came back again the following week to actually scan the piece we had decided on for the set project. I found a few options including a wall hanging plate decoration, a gold-plated frog skeleton, and the object I chose: the marble lamp seen below. For the 3d scanning process I utilized a Canon EOS and a web-based program called Recap 360 by AutoCAD. Throughout the project I used Blender, Meshmixer, Solidworks, Inkscape, Adobe Illustrator, and Netfab for 3-dimensional and 2-dimensional manipulation.

The original scan of the lamp from Baker Furniture only took a couple tries and ended up coming out rather well. The shadows caused trouble when attempting to scan the object. The scan took about 40 pictures in total to create. The pictures had to be taken from as many angles as possible around the lamp.

After cleaning up the original scan I ended up with this model above. This form was brainstormed as much as possible during the first week of the project.

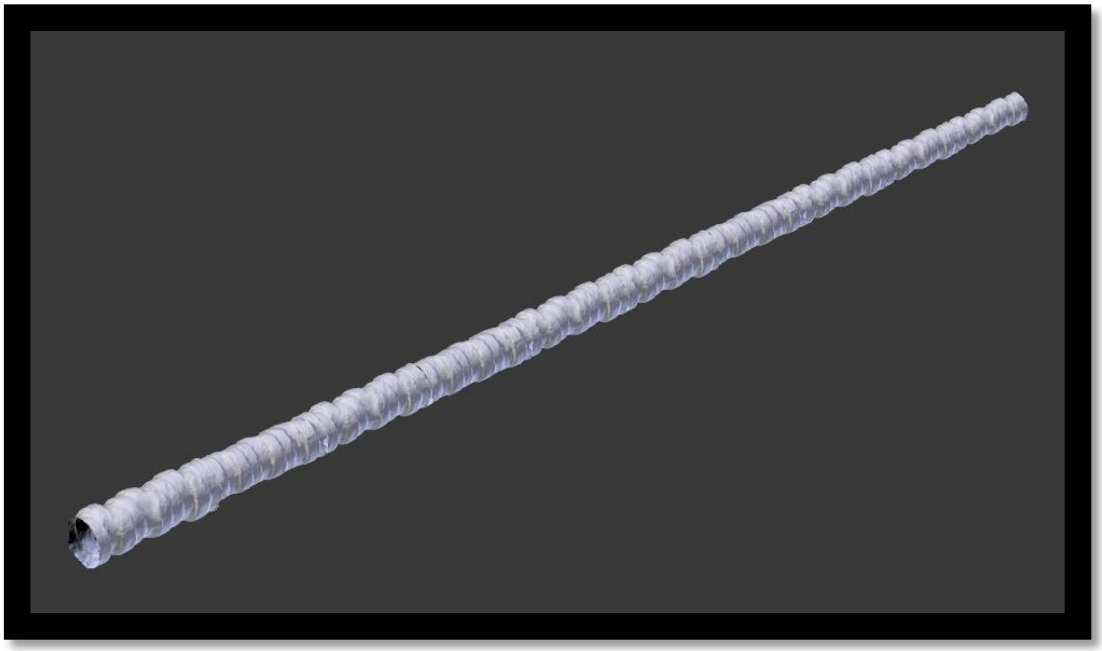


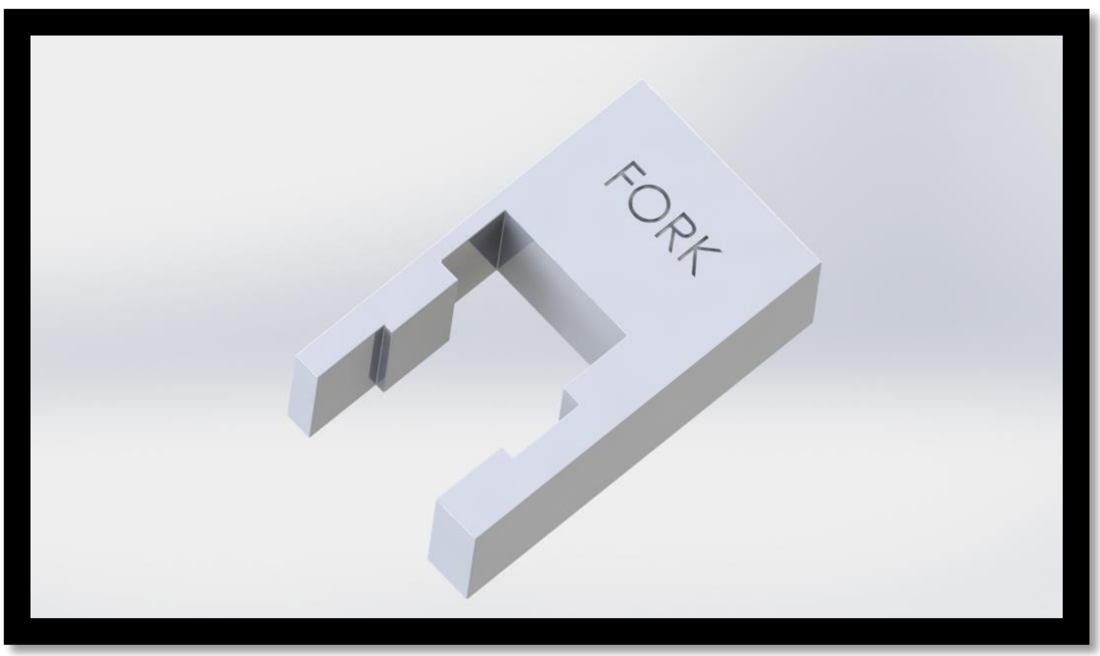
I then took that model, scaled it, and then proceeded to repeat the form and join them end to end to form the beginning of the chopsticks' design. I was working towards a design similar to the mock-up displayed below.



Week 2

The second week I created a model with the correct length, taper amount, and cleaner joints between the each separate model used to make up the entire length of the chopstick. This model is the most basic, unsolidified, version of the base and chopstick as well as the other design elements taken for use on the spoon, fork, and knife heads as well as the outlines used for the top of the box for the product set.

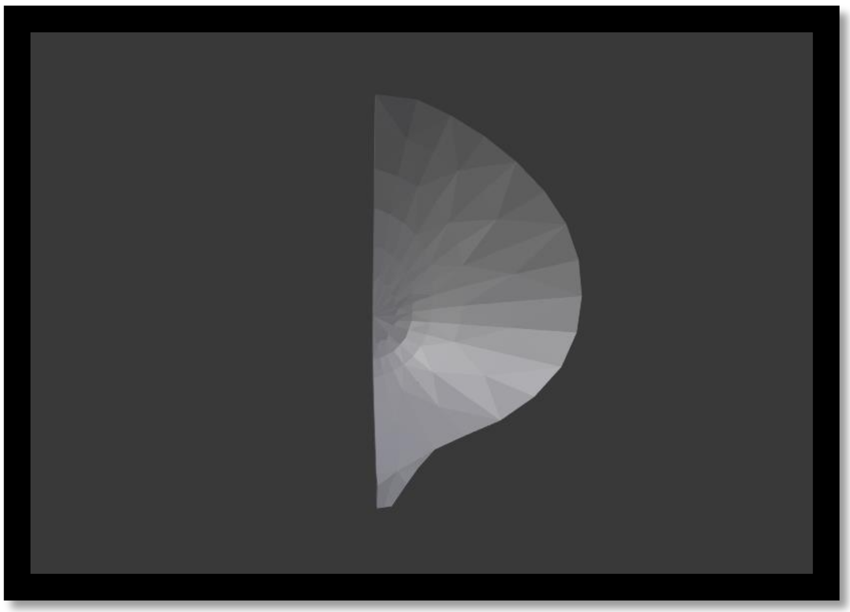




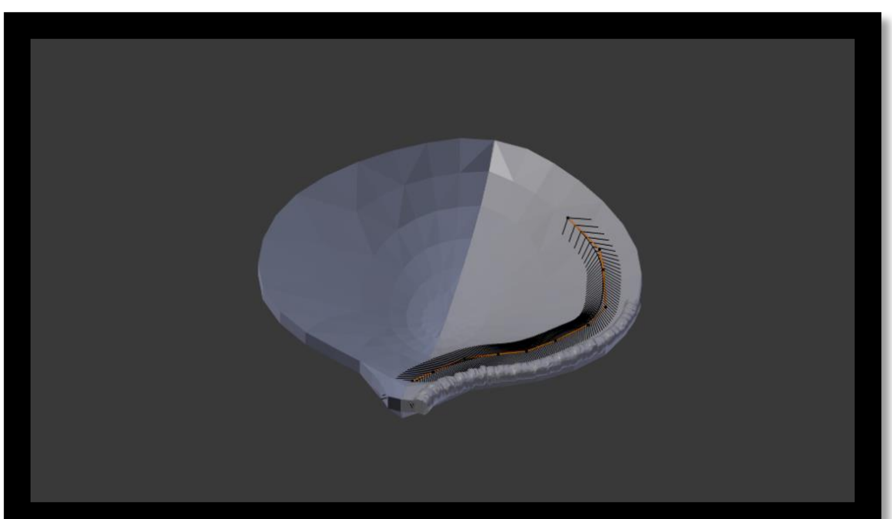
The above model depicts a possible form of connection between the base handle and the various attachments available. This idea consists of using the form of the end of the base handle as a key inserted into the keyhole portion of the attachment. A fork is shown with this attachment style above. This method of connection was later discarded for a more stable and forward system. Join us next week when I continue on to create a proper method of connection.

Week 3

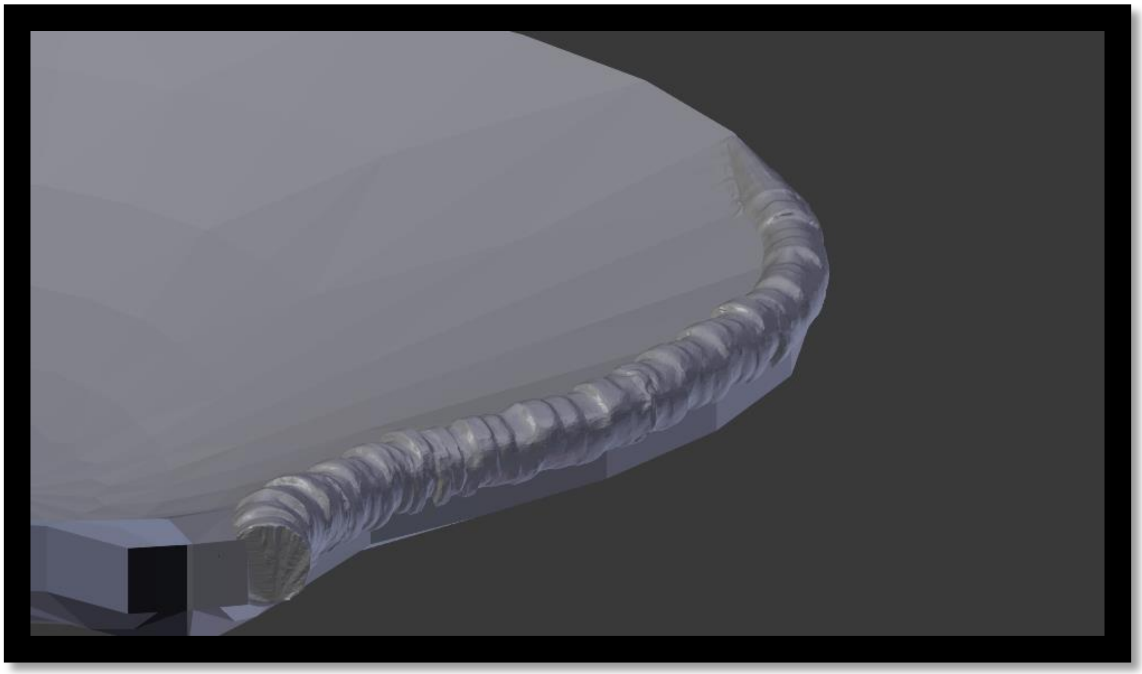
The third week of the product set project involved solving the connections issue, continuing arduous 3D modeling, and continuing to brainstorm surrounding ideas. Some surrounding ideas including the nascence of a wooden box to hold the whole set as well as tie every piece together from a design point of view. The most attention was paid towards creating the spoon attachment and refining it a bit. I began the spoon by making only half and seeing which shapes I liked and then mirroring the half spoon. For example:



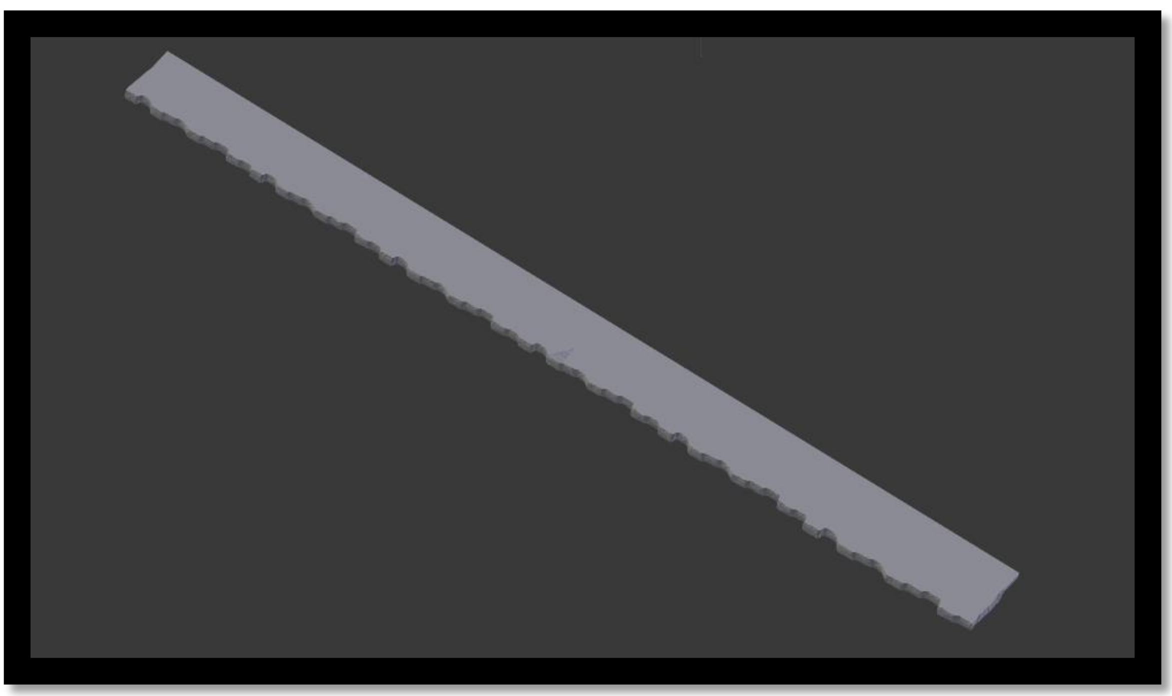
After creating two or three of these I decided on one design and went on to refine it throughout the week. At the end of the week I had a poor rendition of the soon head that you can see below.



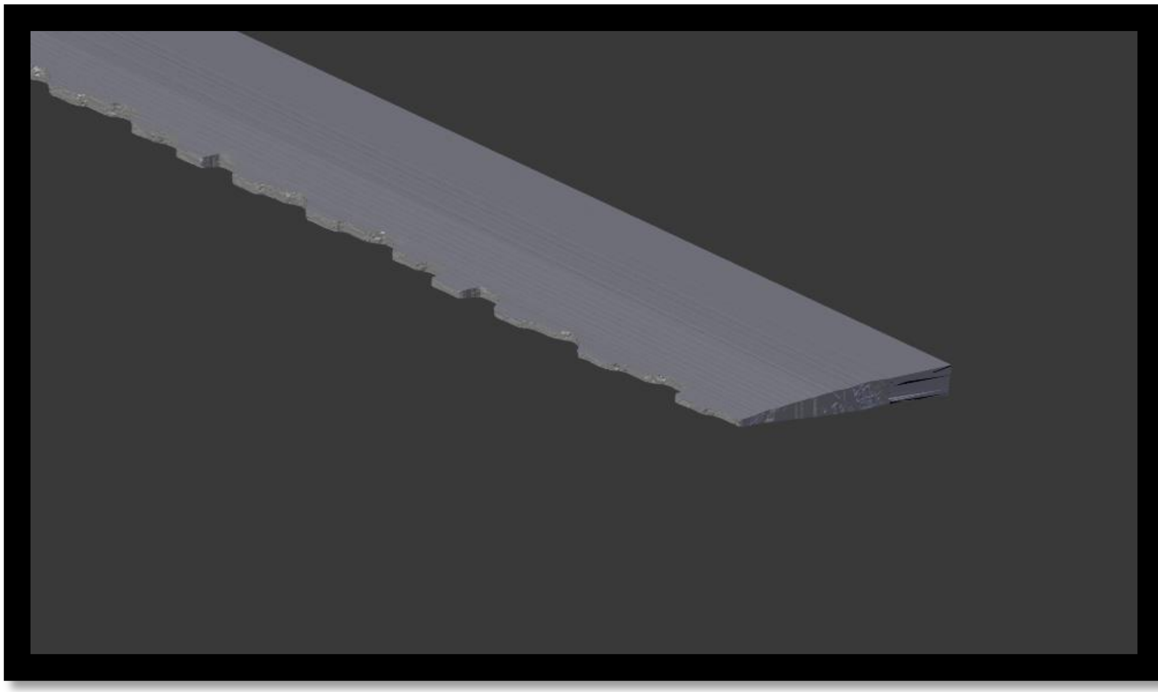
You can also see the path used to guide the position of the column design element that ties the set together. Here is the model a day later after joining the objects and smoothing them together:



Next, I worked on the knife a little until I got stuck by my lack of 3D modeling skills. The knife's edge is created out of the shape of the column this week. I cut the column until I got a shape such as this one:

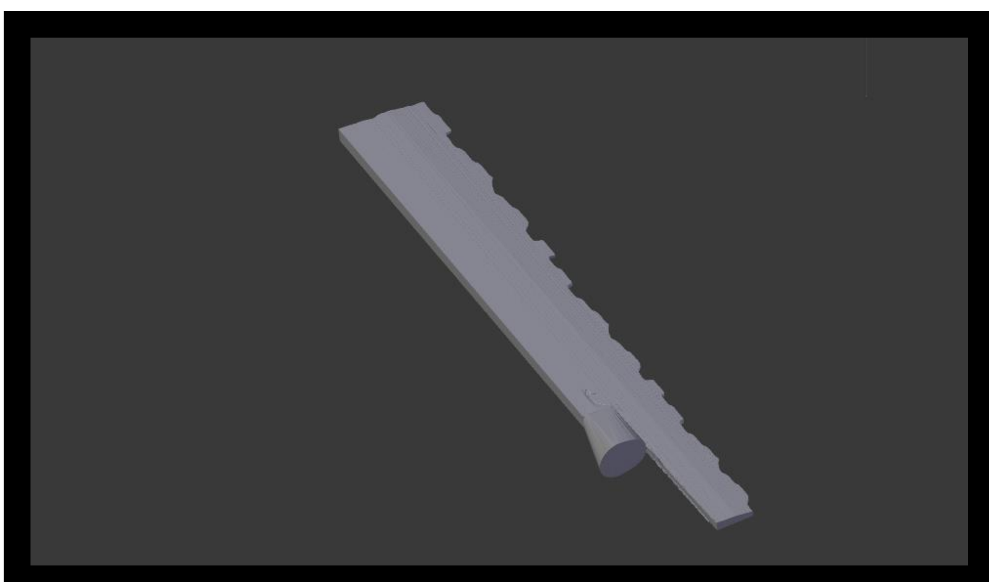


I continued to transform the bar to achieve a sharper cutting edge like so:

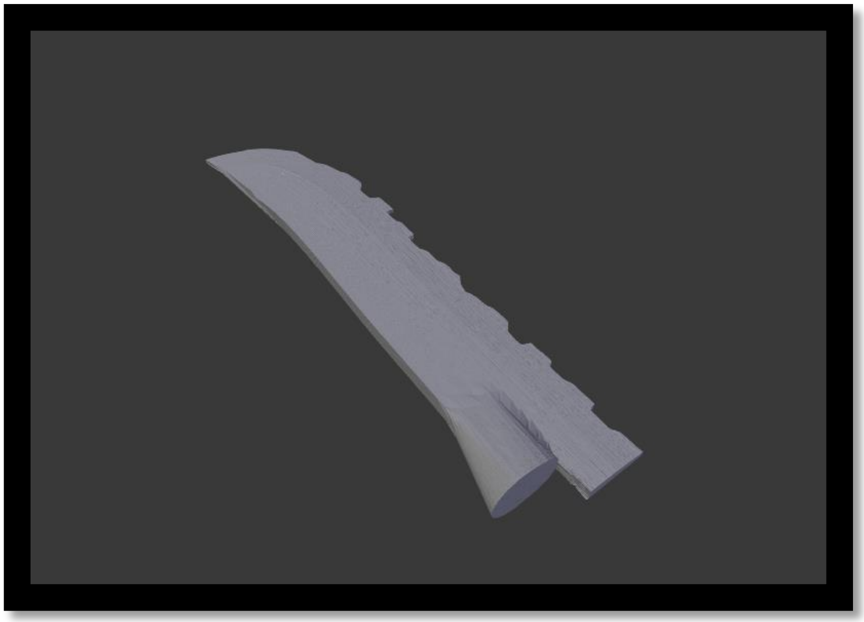


Week 4

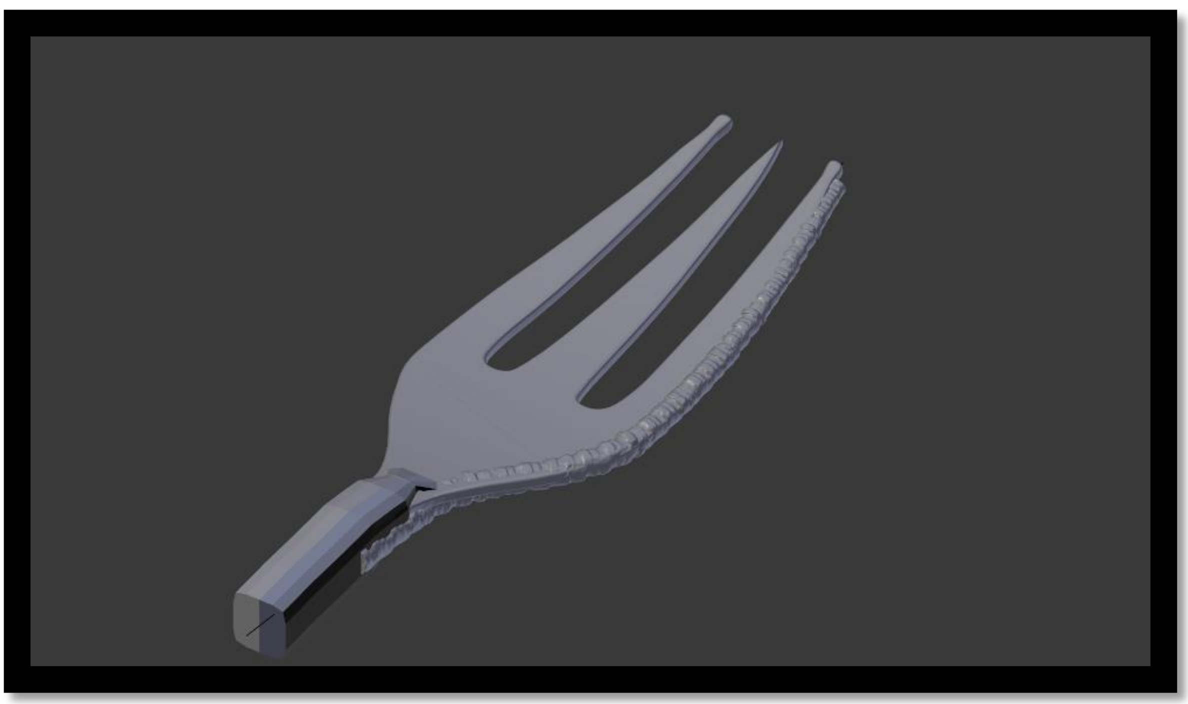
To start off the fourth week of the project I continued on the model of the knife and began to design a matching fork head as well. First off, the knife grew a way to connect to the base handle. I decided to leave the “tail” feature just in case I wanted to keep any of it later.



Then I spent at least two hours trying to find out how to bend the piece into a curved blade shape. Finally I was able to find out how to use the correct modifier on the object using Blender. I found throughout the project that it was sometimes more of a crash course in 3D modeling than anything else.



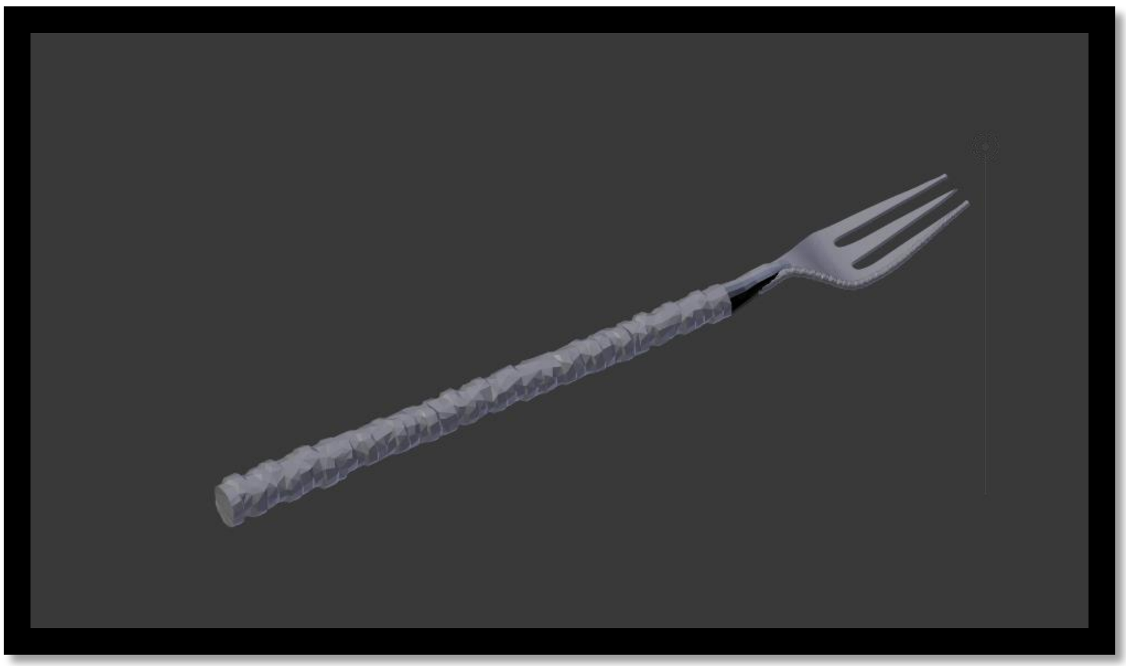
I also ended up shortening the extension at the base of the knife head. This is the final model of the knife head right above. It actually turned out exactly like I wanted, even if it did take an unnecessary amount of time. Then I moved on to create the fork head.



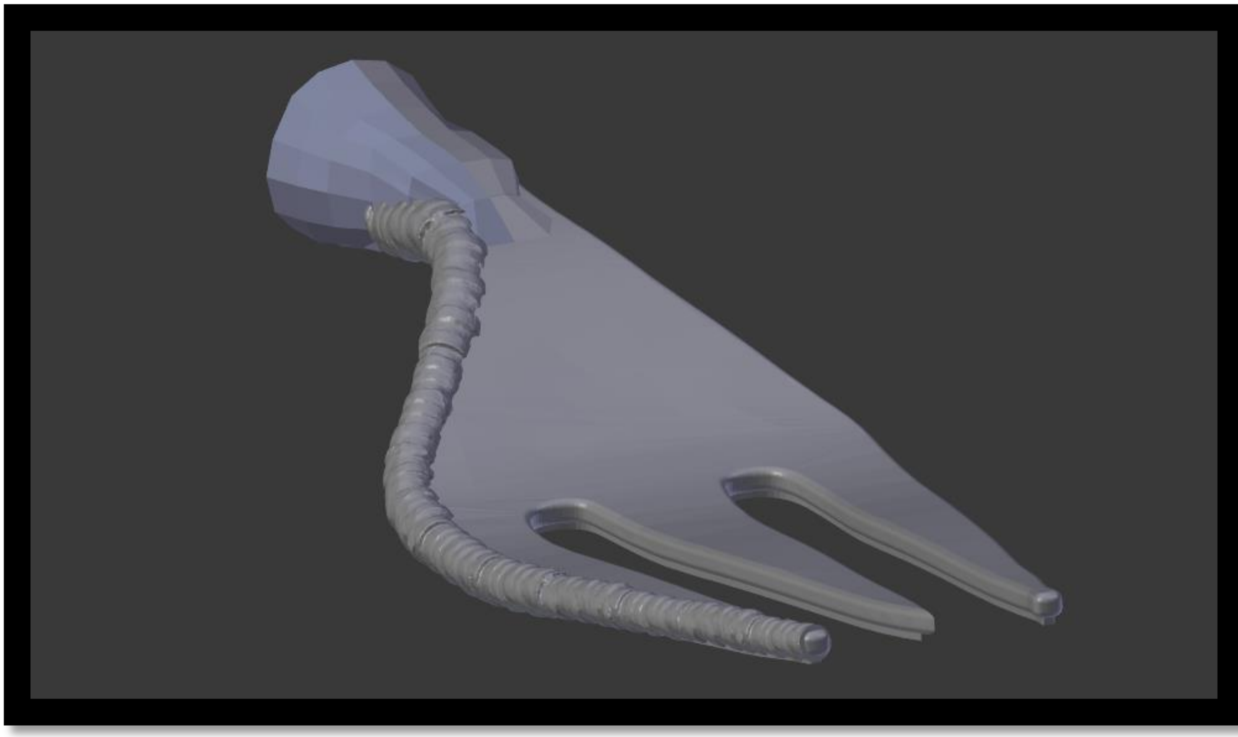
I had the idea to create each of the prongs out of the scanned form but decided to go with the melded sort of look that I got with the spoon head. I again used a path to position the design along the fork's prong.

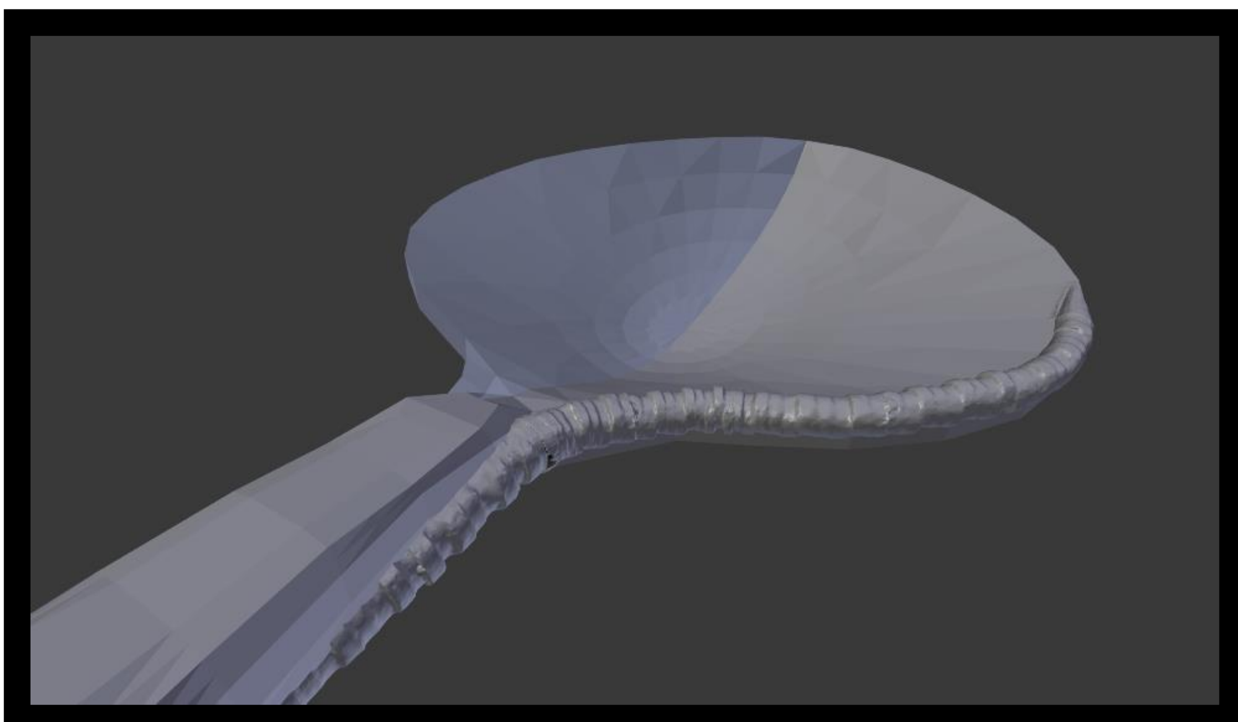
Week 5

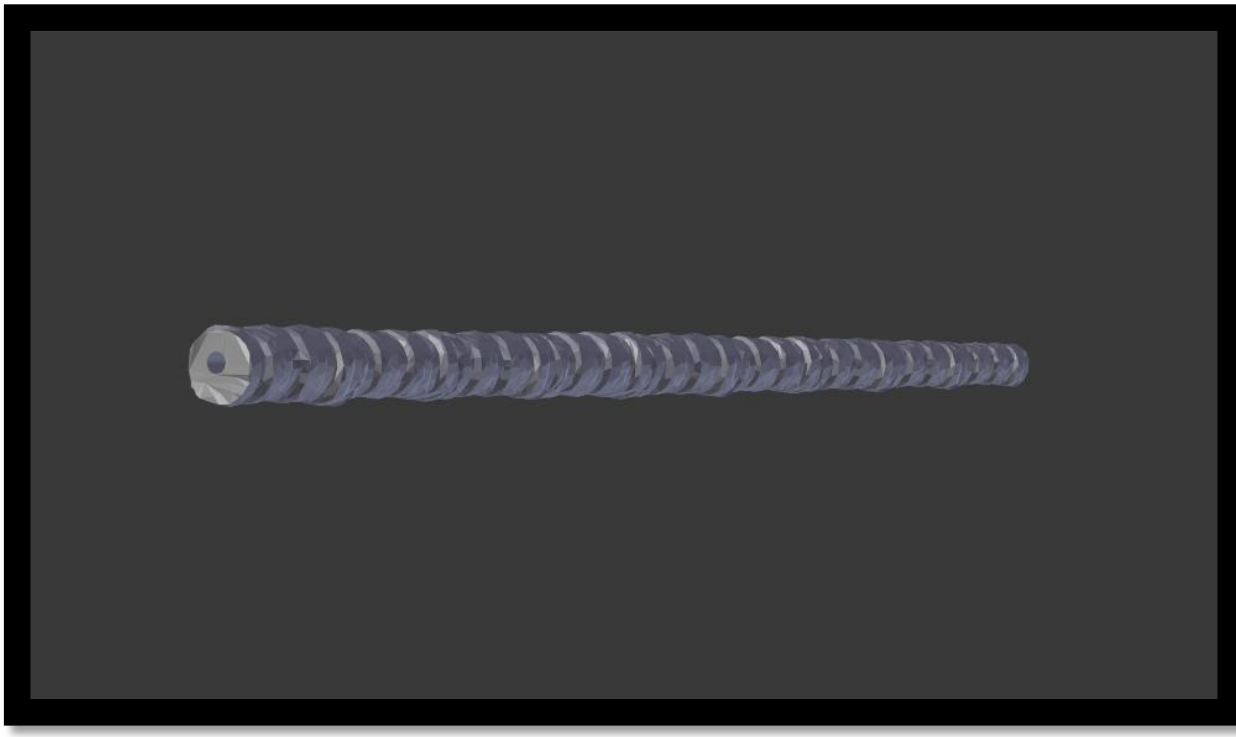
First off, I finished polishing the fork and spoon attachments along with sizing said attachments to the base; especially focusing on the connection area of the attachment. Here is the correctly sized fork for example:



Here are the attachments in greater detail in their final form:

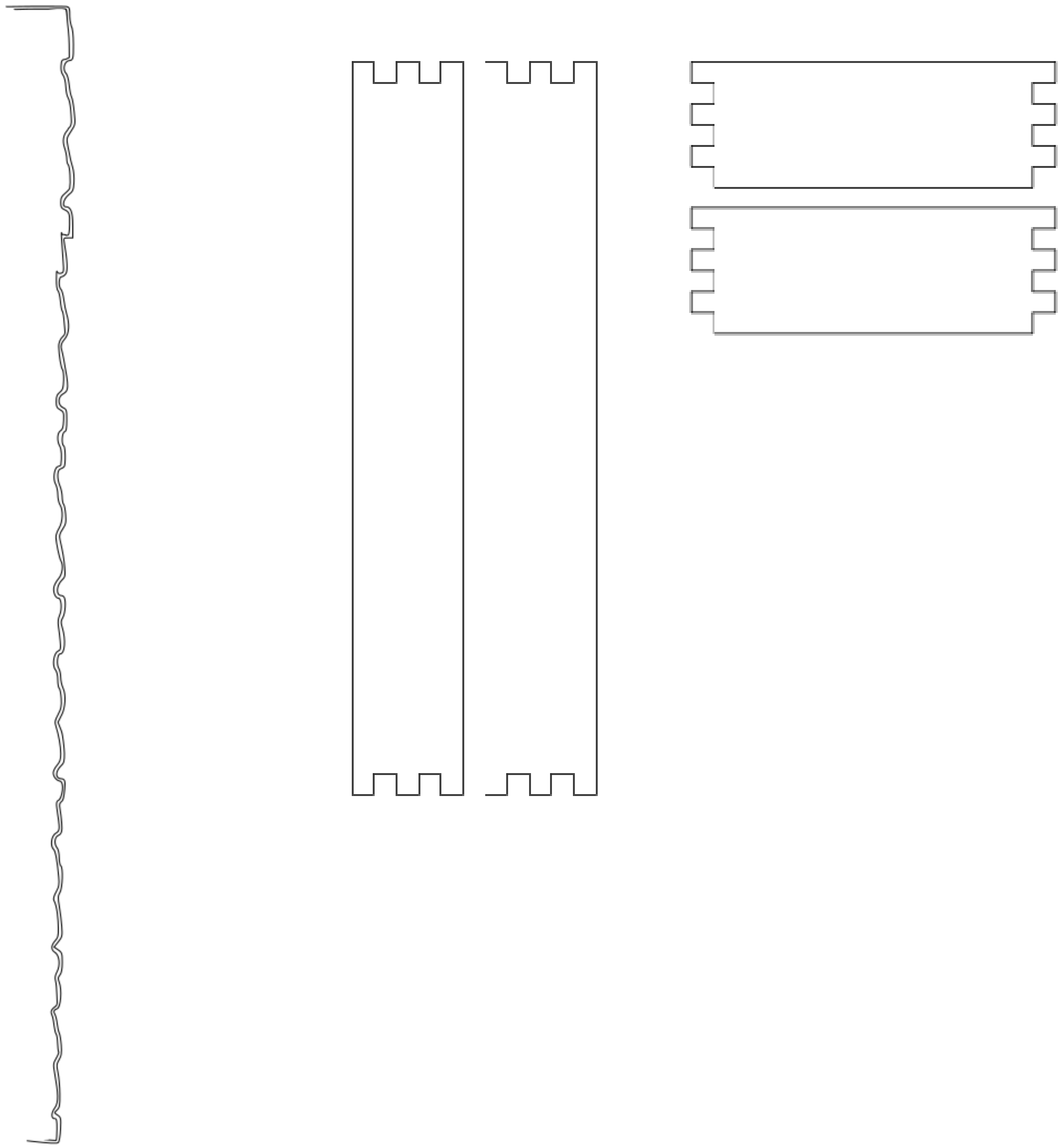




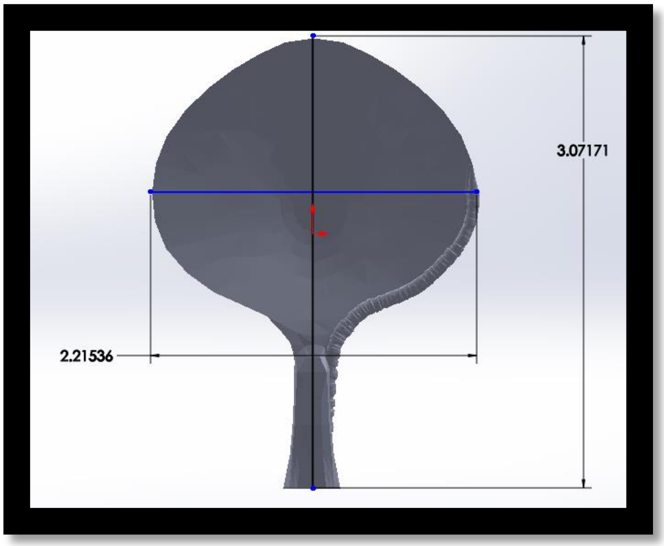


During this week I also finalized the ornamentation. The ornamentation is the hollow tube that is shown above. It allows you to drink and eat your soup at the same time. It is a functional ornamentation that adds a new functionality to the object. A secondary ornamentation was also decided upon. I never got the chance to actually make the whole box; however, the bottom of the box would have a small cloth for aesthetic and functional purposes. It would keep the ends of the base handles that are being placed against your lips in order to sip the liquid through. This ornamentation would also be very aesthetically oriented since it will be the only thing sprucing up the inside of the box other than the pieces of the set itself.

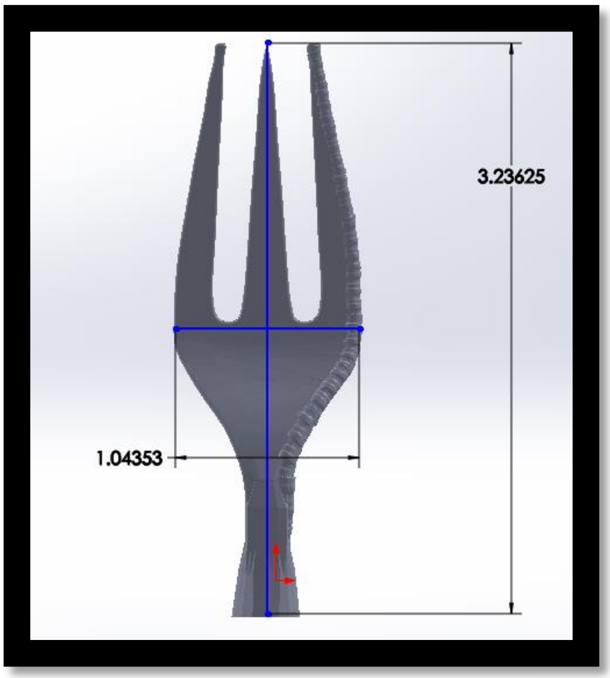
Along the path to creating the design on the top of the theoretical wooden box I had to create the best possible outline of the form seen above. I did so by using this picture in Inkscape to trace out an outline. I created the top of the box to have only the design etched into it like:

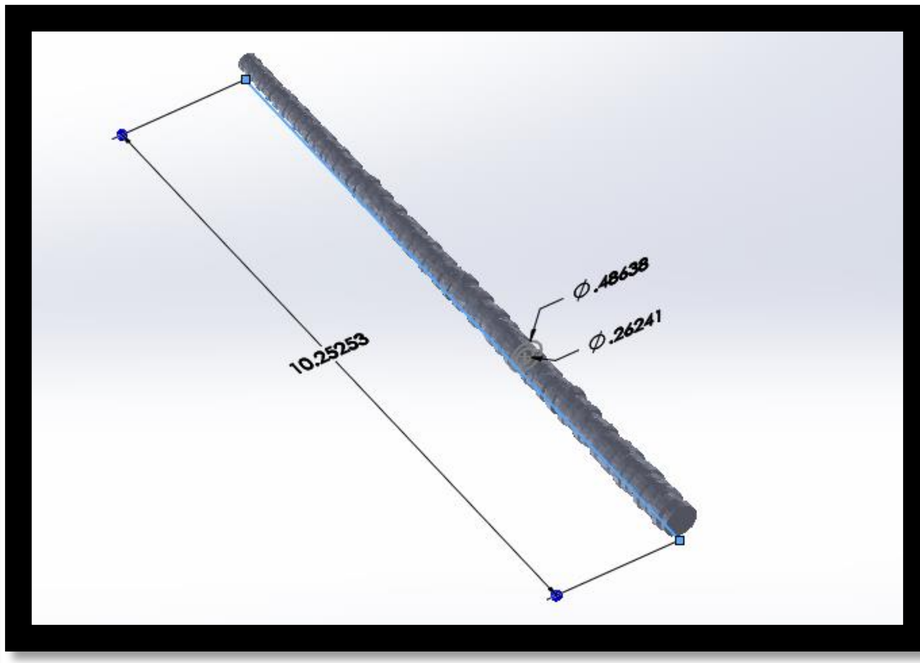


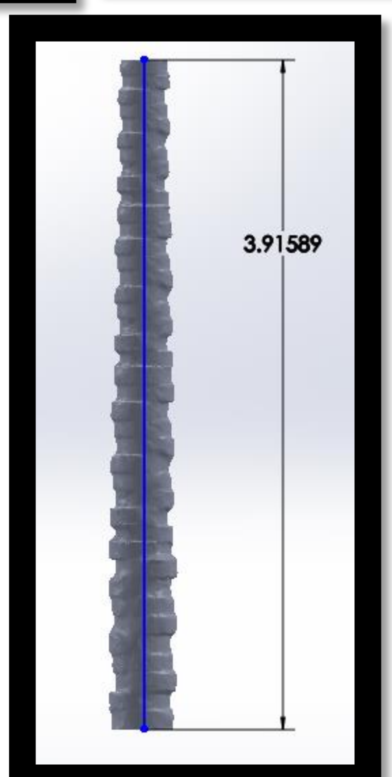
These are the four vertical sides of the box. The connections are made by finger joints that are to be glued once connected.

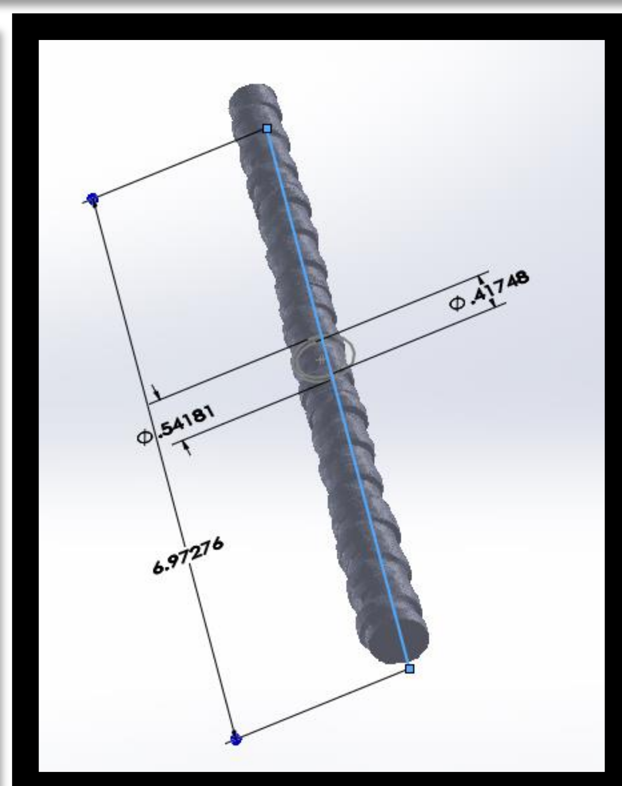


Dimensions of the different
components of the set

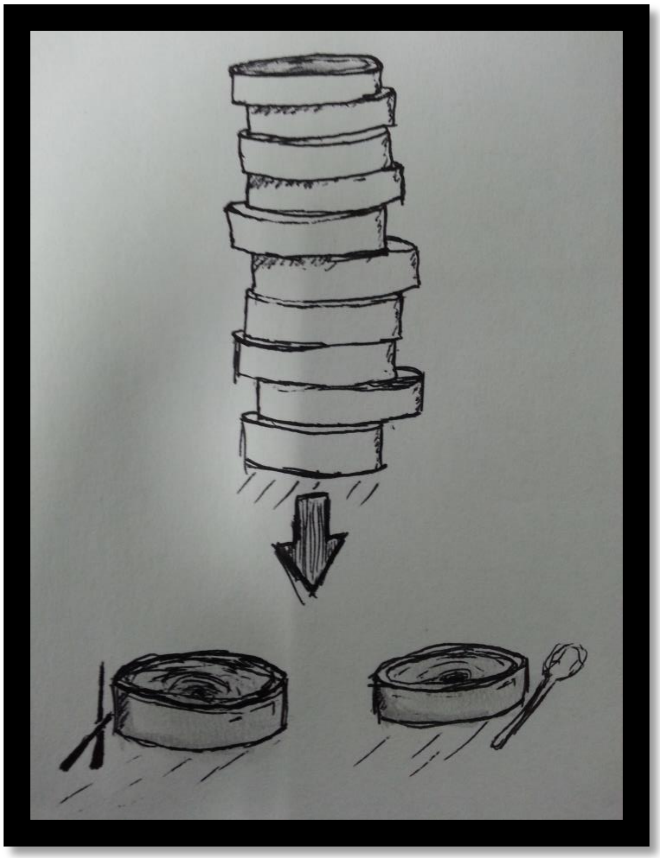


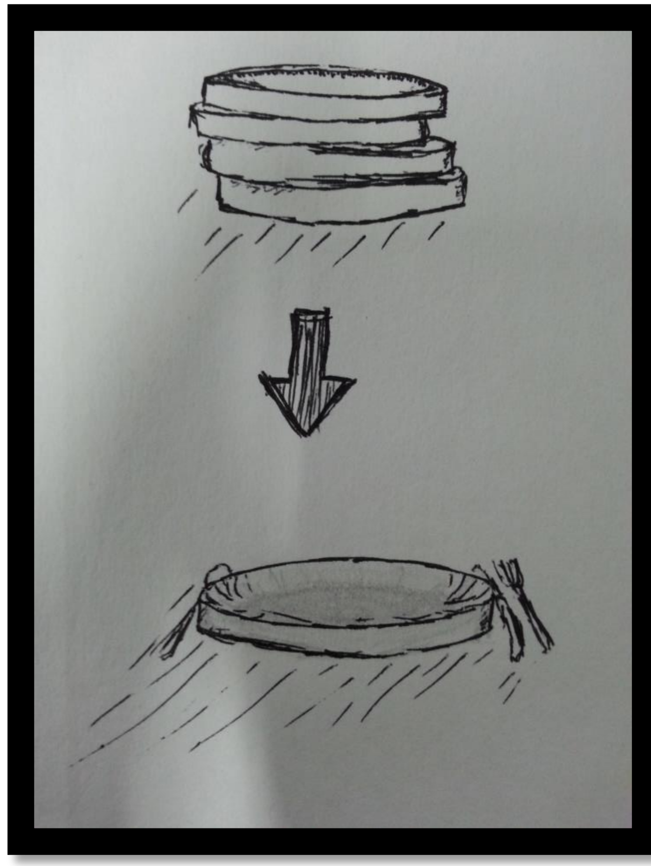


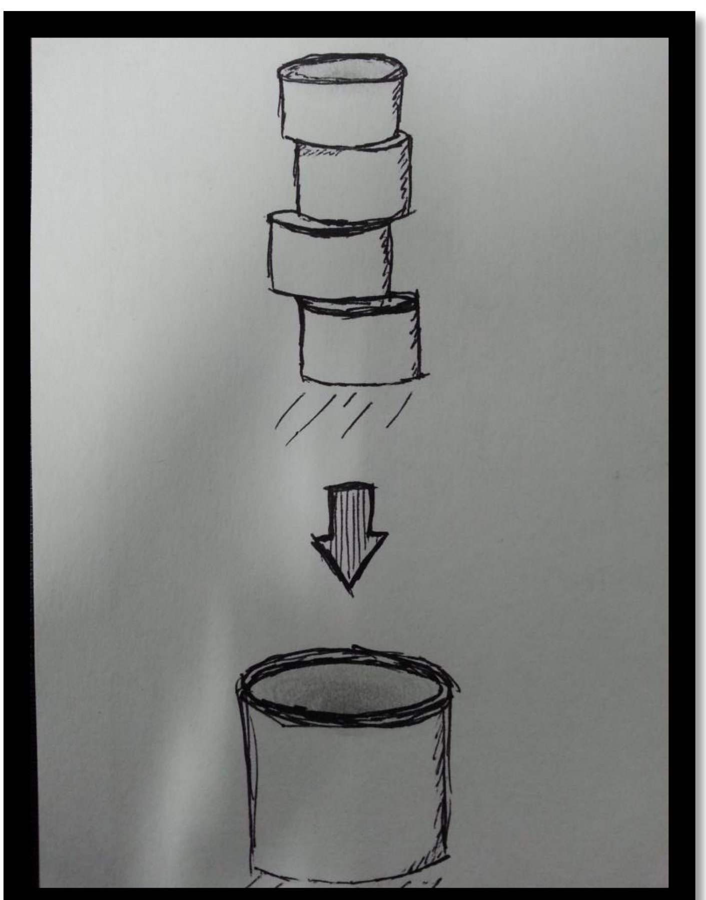




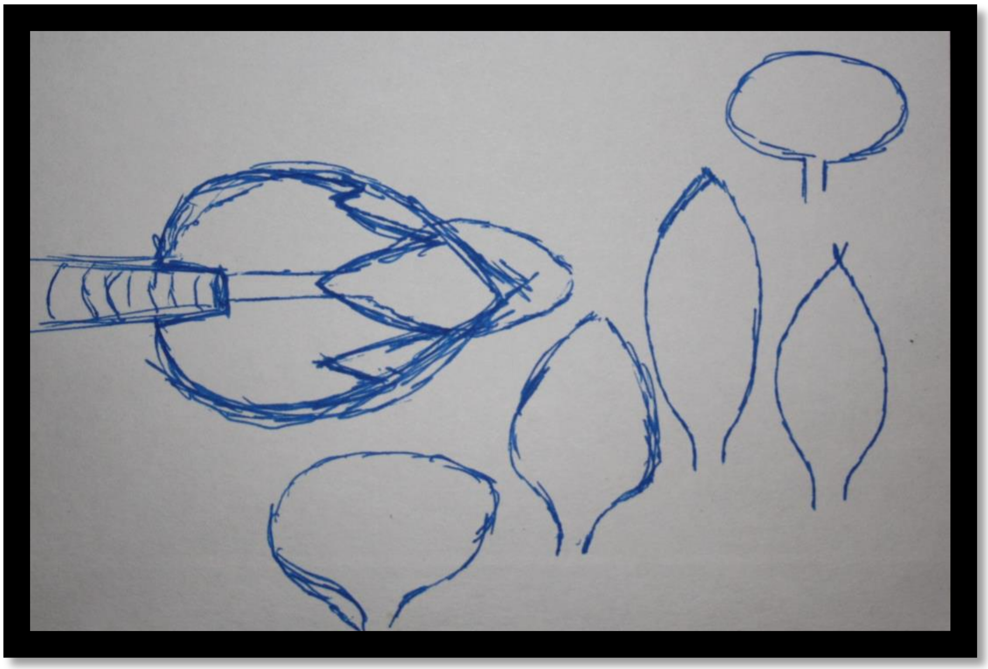
Support Materials



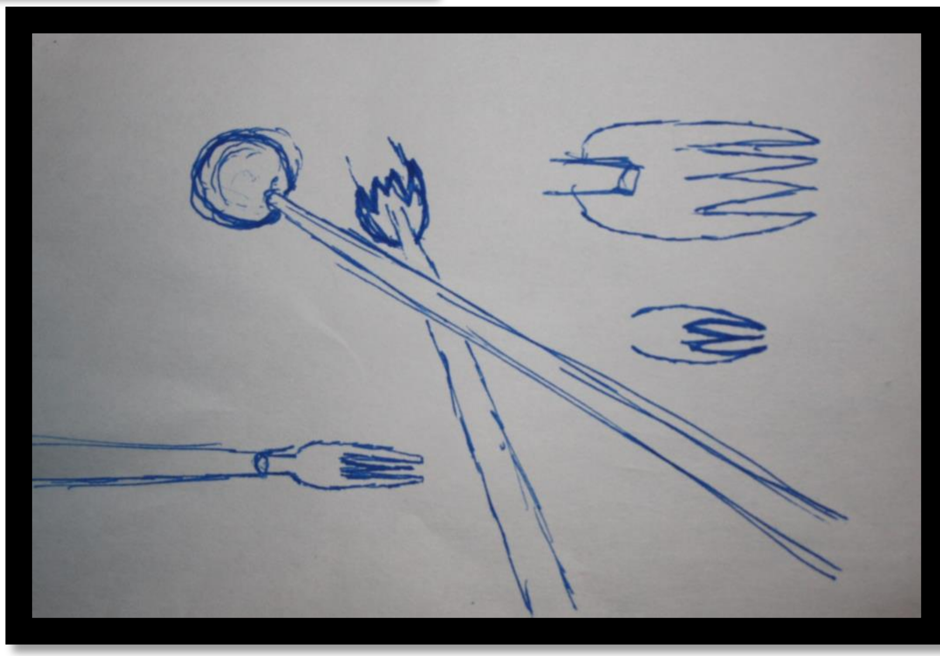


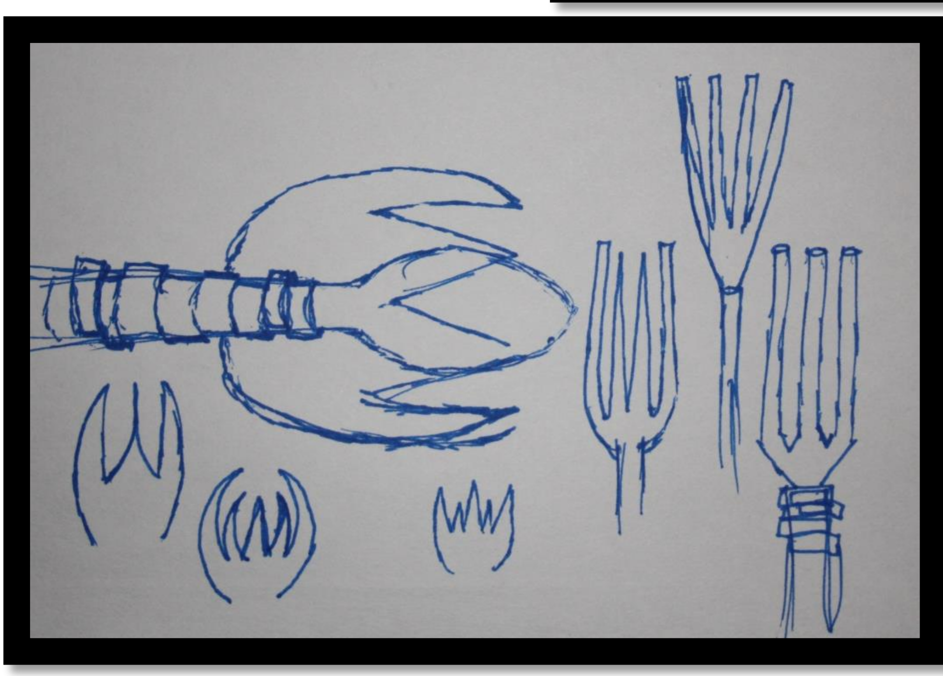


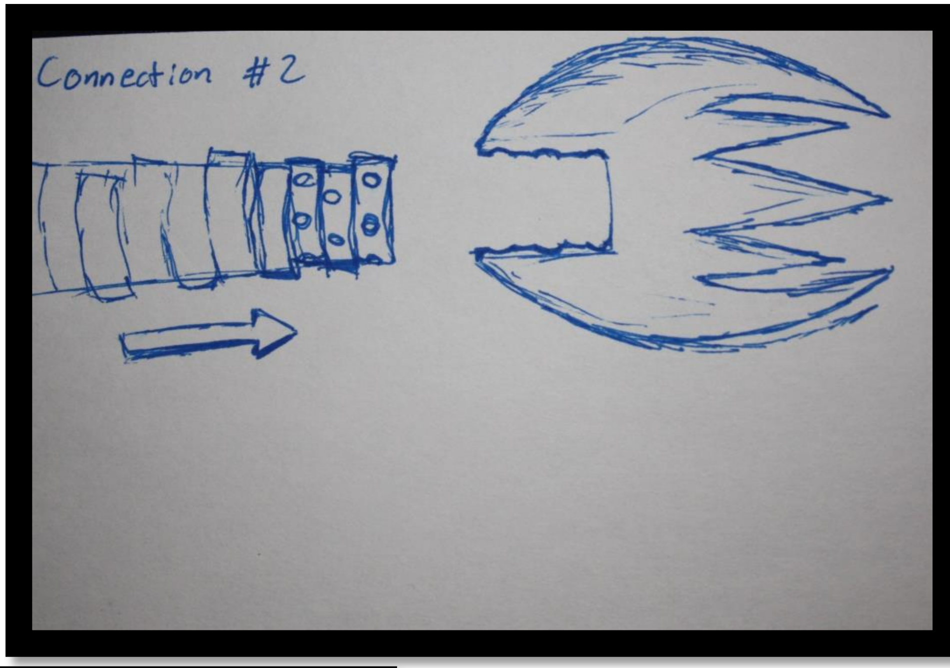
The preliminary idea for the product set.
Stackable bowls, plates, and cups



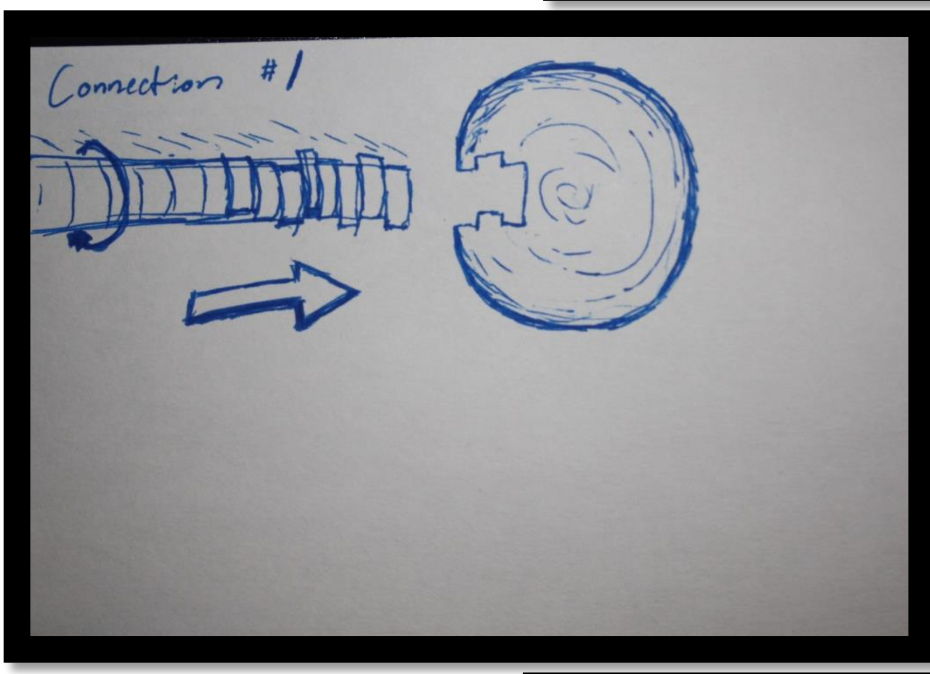
Brainstorming the shapes of the different attachments

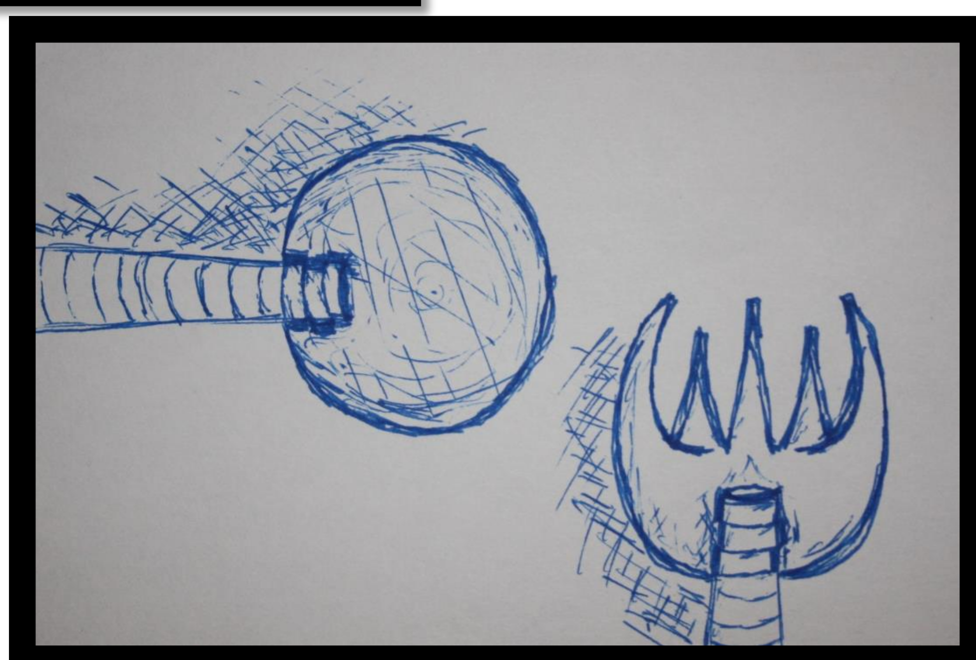


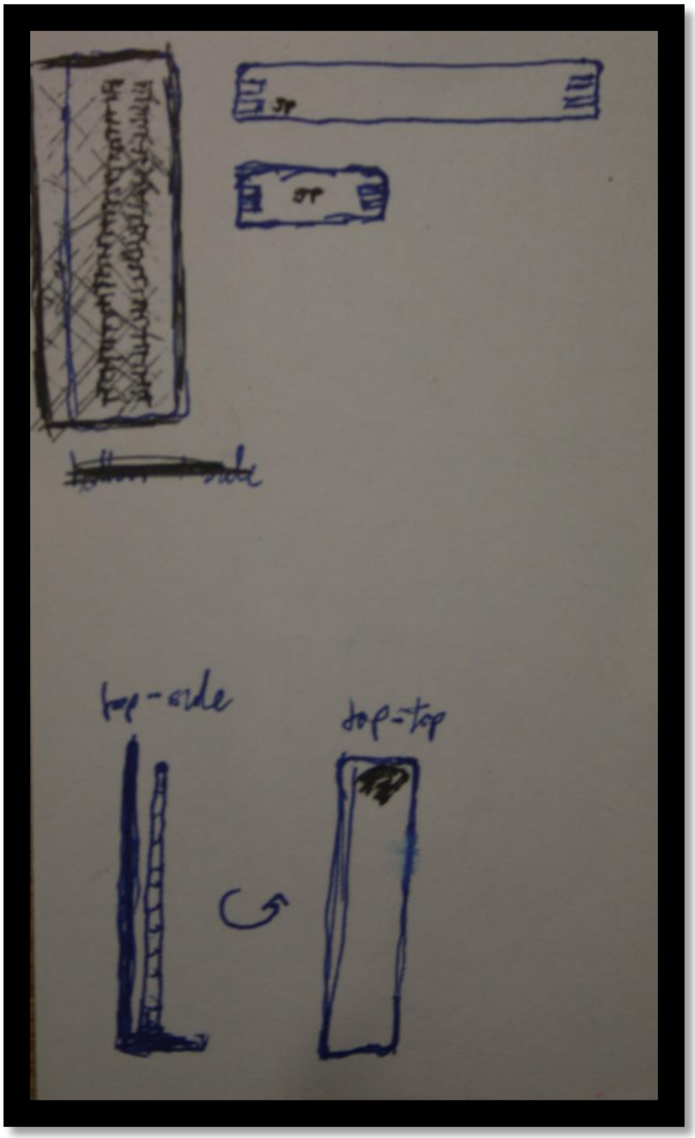


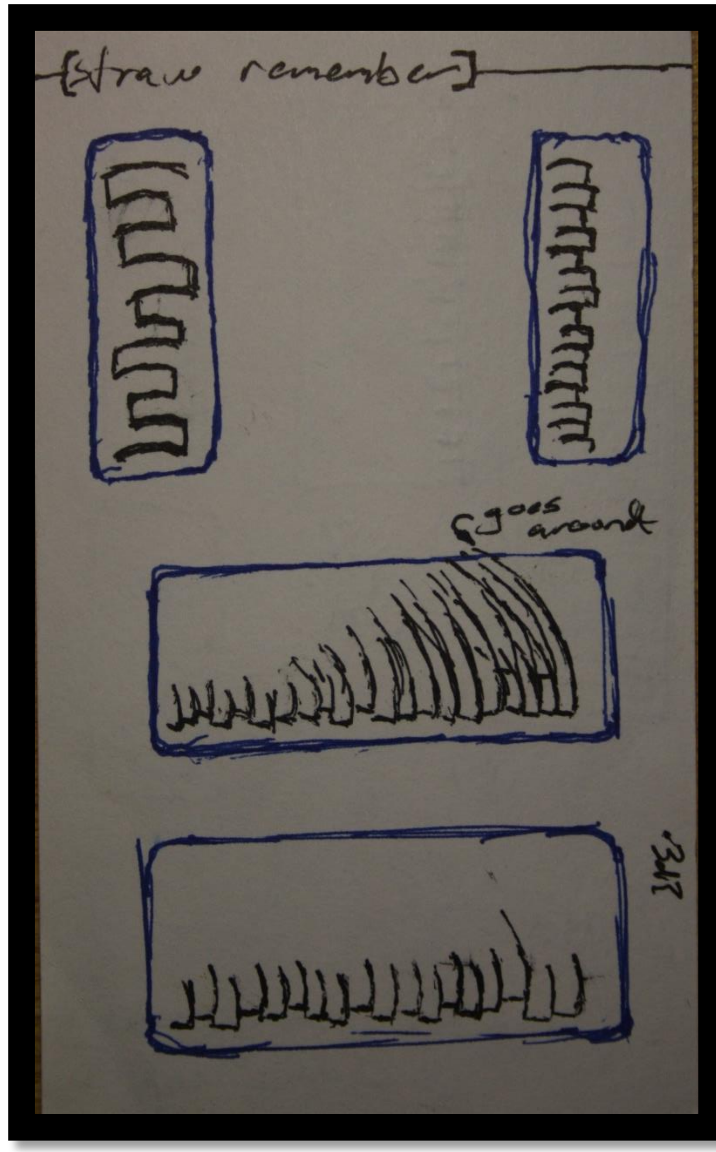


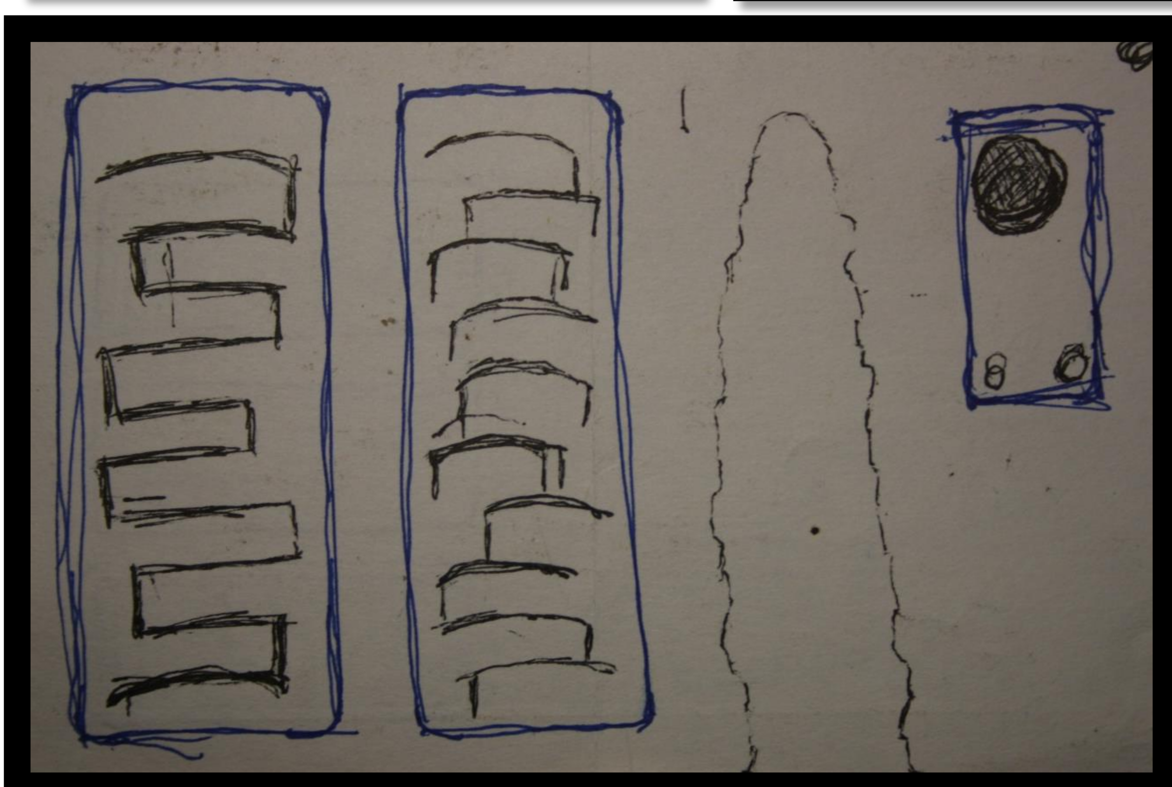
Two of the many ideas for how to connect the attachments



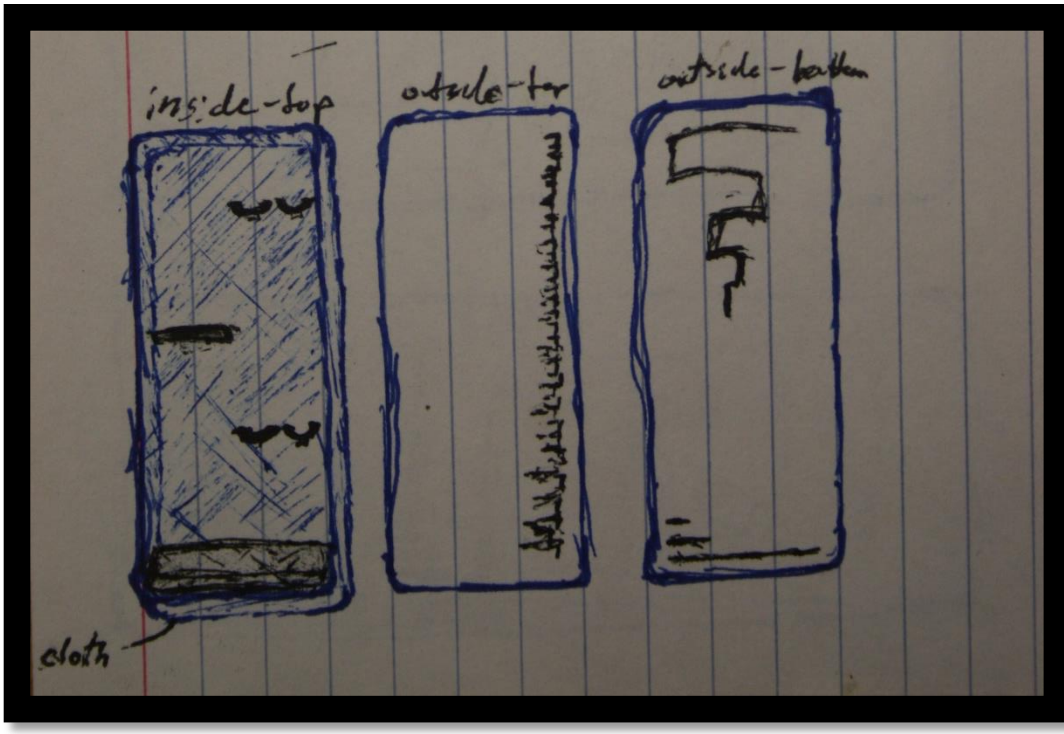


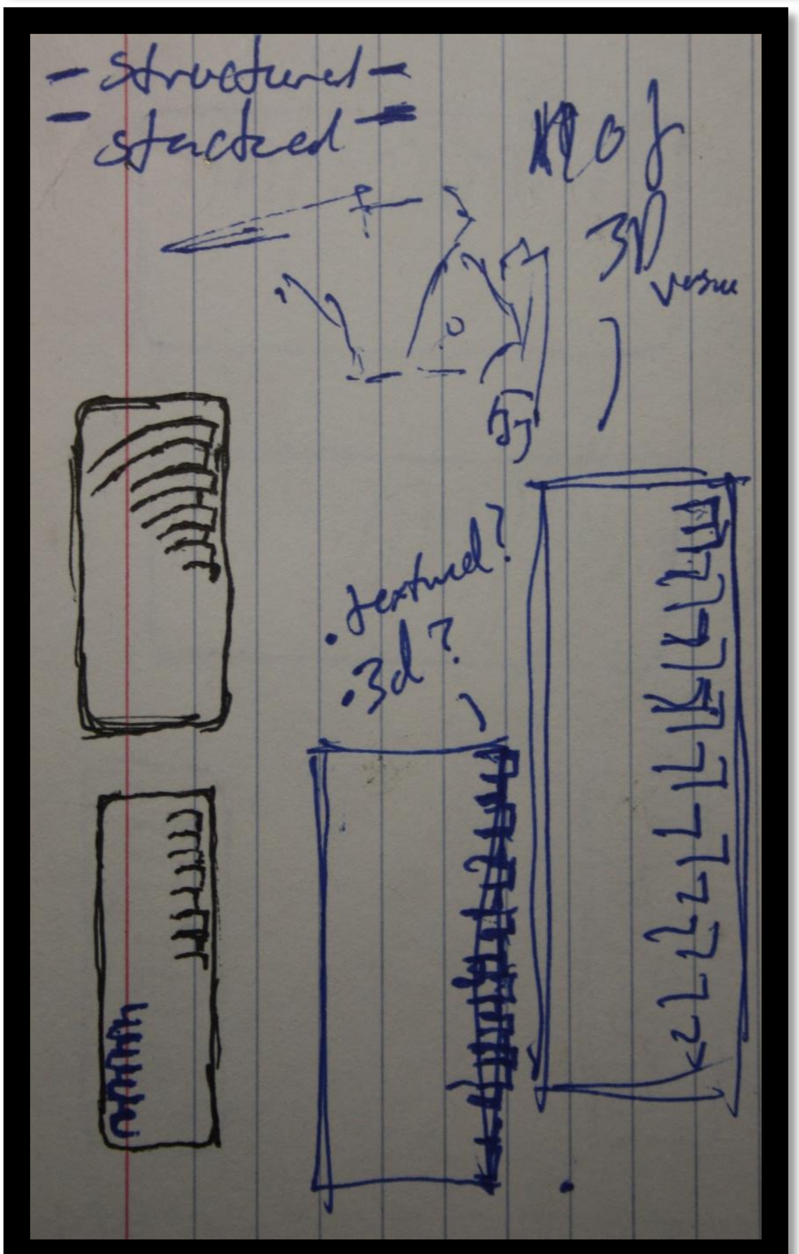




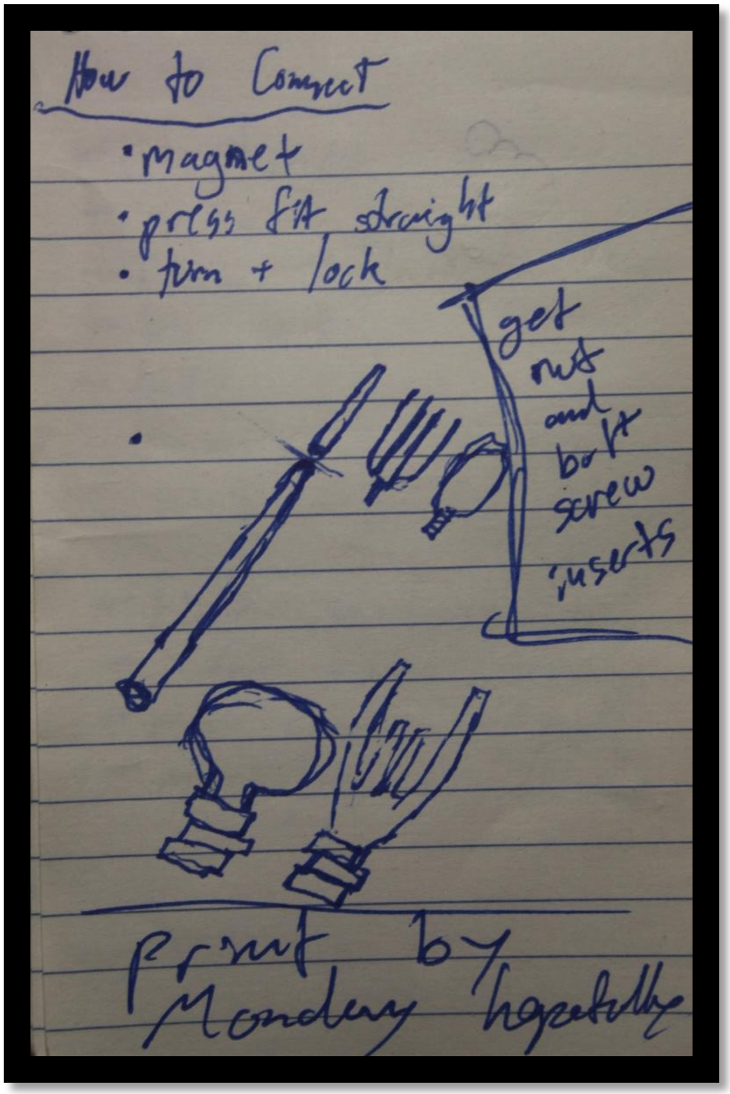


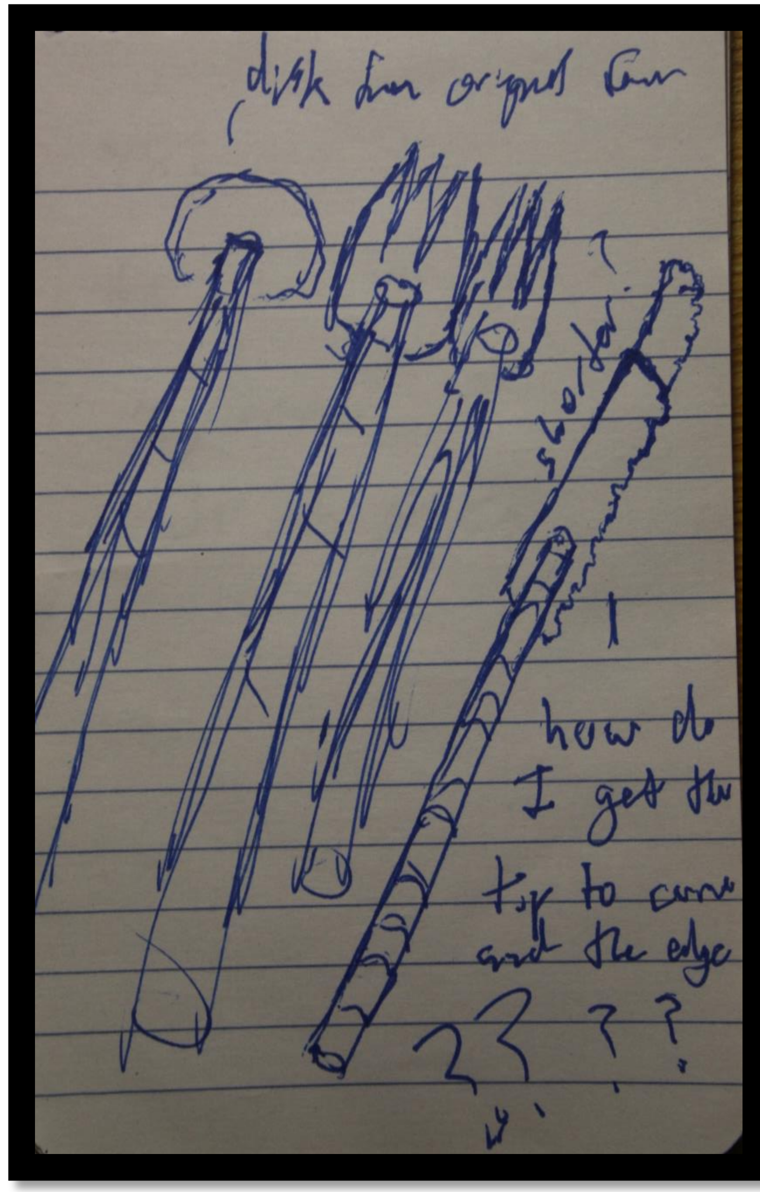
Sketches detailing a few ideas regarding the box to contain the utensil set, especially the top of the cover



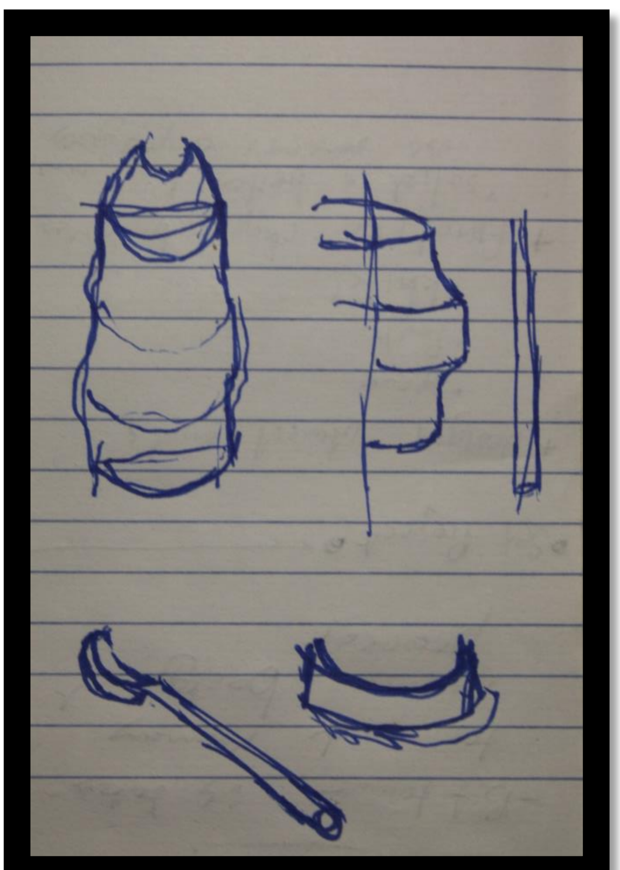


Sketches of the multiple sides of the box continued





Quick annotations and miscellaneous sketches



Spoon cutouts that I attempted to use to create the spoon bowl section but failed. The sections are all dissected from the original scan.

Ornamentation

The ornamentation chosen for this product set was a hollow tube piercing the axial center of the base handle and chopstick tips as well as six perpendicular holes near the tip of the chopstick. These tubes are function focused. This design element makes it possible to sip your soup and grab your noodles with only one tool in hand. The comprehension of use of the straw is aided by the holes near the tip; otherwise it may be hard to see it is also meant to be used like a straw. A hint of mystery is added to the aesthetics of the product since it is not expected to have that functionality tucked away inside.

Smart Integration

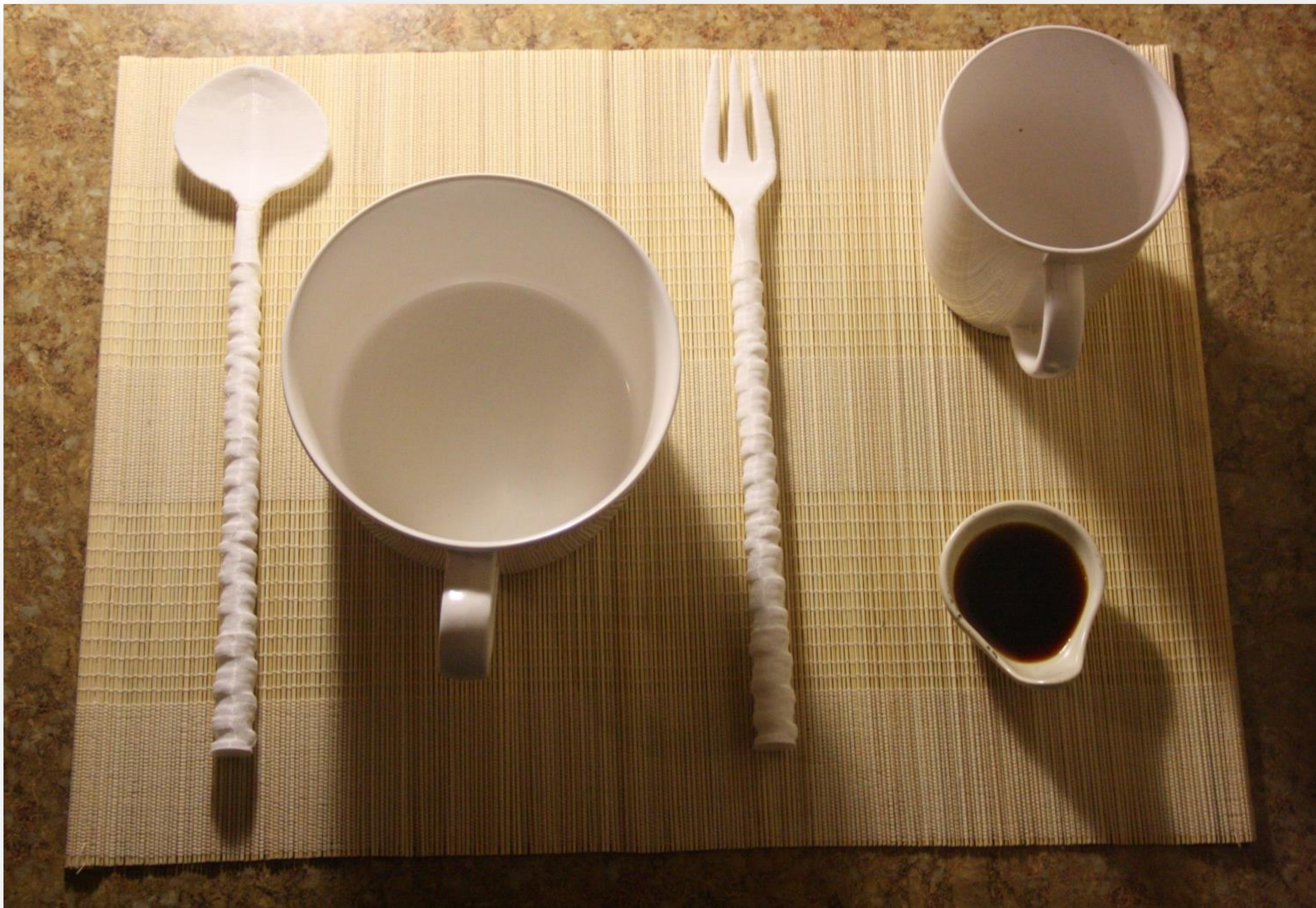
The electronic integration for this set gives the utensils the ability to make whatever spoonful, forkful, or mouthful of food you eat have the perfect amount of salt, or pepper, or maybe even Sriracha. The device was designed to be housed inside of the base handle. It uses a motor and plunger system to release precise amounts of whatever it is you want n your food. The tip of the utensil will be able to dispense a preset amount of flavoring substance at the press of a button. The system is also designed to be used with medical patients who would benefit from an easy medicine dispensing system that goes straight into your meal as you eat it. This avoids the displeasure of taking medicine, increases ease of use, and also encourages the healthiest way to take medicine. Both functions benefit from the integration of the tube running through the middle of the chopsticks.

Thesis

Physical objects are now completely malleable when electronically translated between the computer and the real world. One of the most useful aspects is being able to add a previously unrealized functionality to an already existing object simply by scanning and then proceeding to edit its form on a computer. For example, I can scan a coffee mug, design a new handle that I prefer, print this handle or whole mug, and now I have the same mug, only with added functionality.

The application of this idea is one of the most exciting aspects of 3D printing today. Being able to change what something does even after it has been designed, produced, and sold changes the basic ideas we have about objects in the physical world. Previously we've always had to start from something else to change a physical object in such a manner. We wouldn't be able to take a ceramic flower vase, pinch here, move it a bit there, and come out with a new transformed vase. Now, with the use of computer assisted scanning and modeling, we can. After translating the physical form into the computer's language we can pinch and pull however we like and then translate it right back out again with the help of 3D printing technology. Imagine just scanning your pen and printing out a similar one but with two tips, or maybe you print out an eraser holster to attach to it. The opportunities are now much more endless than before.

Photographs



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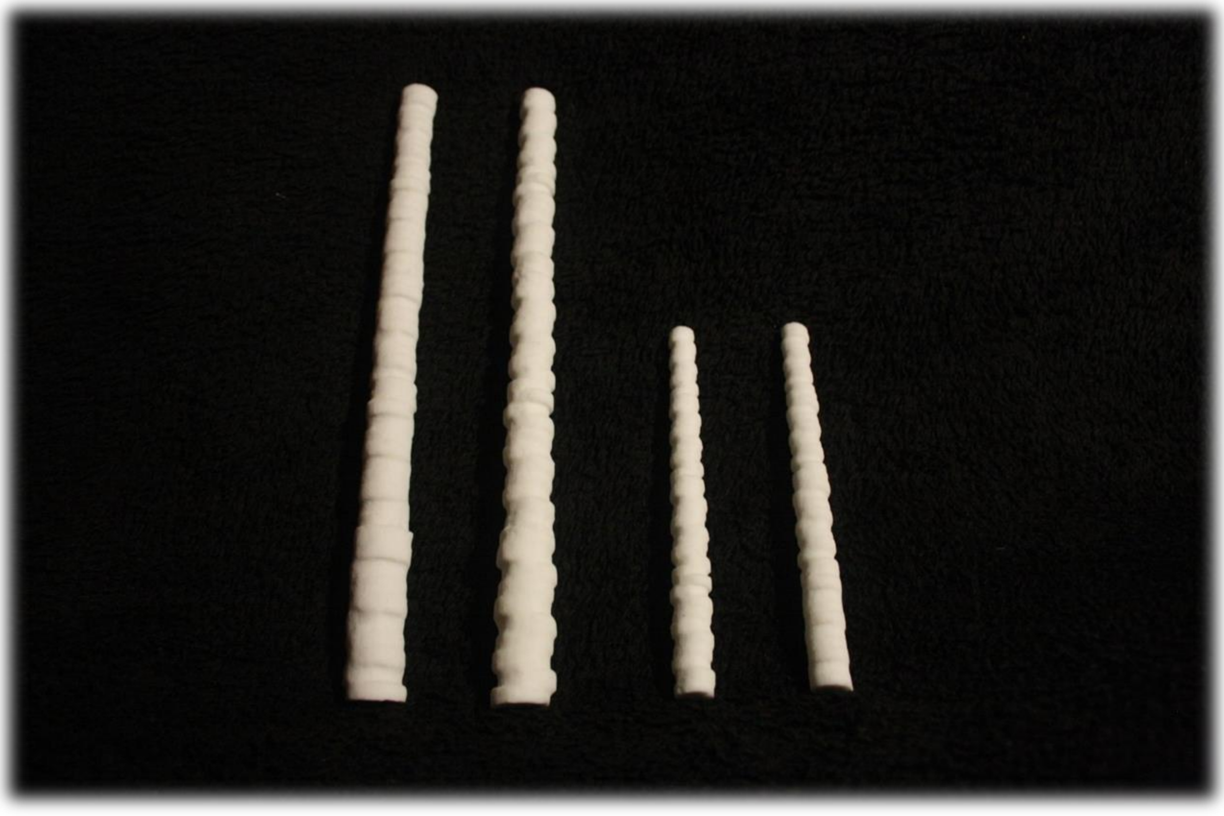

Instructions for Use

Operating Your Utensils:

1) -Position your hand like you're holding an egg.
   -Have the board side of the chopstick lay on the connection area between your thumb and your index finger.
   -Use the top part of your ring finger, your thumb knuckle and the connect area between your thumb and your index finger to perform like a lever. Don't hold it too hard.
2) -Keep your hand in the same position as step 1.
   -Use the top part of your middle finger, the lower part of your index finger and the top part of thumb to perform as a lever. Don't hold it hard.
3) -Keep your first chopstick in the original position.
   -To close, push your middle finger, and the second chopstick will get close to the first chopstick.
   -To open, loose the strength on the middle finger. The second chopstick lever will automatically make your chopsticks open.
   [Instructions Provided by *How to Use Chopsticks* by alanmon]

Switching Your Utensil Attachments:

1) Simply grip the base handle of the OmniCut utensil while turning the attachment head in a counter-clockwise direction to remove it.
   ATTENTION: Be sure to take care when removing sharp attachments such as knives.
2) After selecting one of our OmniCut designed attachments or one of your own custom creations, firmly grip the base handle and rotate the attachment head in a clockwise direction. Ensure a proper connection to avoid unnecessary damage to your favorite set of tableware.

Utilizing the Electronic Flavor Additive System:

1) Lightly press the button for one second to release the customizable pre-set amount of the desired flavoring.
2) Hold down the button for three seconds to receive a double serving of the selected flavoring.

Refilling Flavorings:

1) Open the bottom end of the base handle by unscrewing the end-cap while being careful not to lose the O-ring.
2) Only refill up to the demarked max volume.
3) While holding the open base handle facing upwards so as to avoid spillage, replace the end-cap and securely fasten it back onto the base handle.

Proper Care:

1) Not dishwasher safe.

2) Be sure to clean the attachment mechanisms (the screw and concave ends) with warm water by hand to ensure sanitary use of OmniCut products. Always avoid getting food or other debris inside the connection point.
3) Clean the flavoring container between each use of a different flavor to avoid staining the chamber. When the chamber is empty simply leave it open when washing to clean.
4) Proper care for your own custom 3d-created utensils will vary on the design of the attachment. It is important to note that special care may be required to keep some more complex designs clean. Keep proper care in mind when designing your utensil attachments.

Next Steps

I intend to keep moving forward with this project in order to see it to completion as much as possible. One of the next steps I plan to take is to focus on polishing the designs to be printed in metal or another material. This step involves finding a material that is actually food safe. Due to the methods used for 3D printing the surfaces of the printed items are littered with pits. These divots act as petri dishes for microbial organisms especially when used for foodstuff. The solution to this problem seems to be a proper coating over whatever material it may be.

Secondly, I plan to optimize the sizes of each piece. During the creation process this semester I was unable to find the perfect size the pieces of my product set should be. I have found the correct length to be about ten inches when used as chopsticks with somewhere between five and seven inches of that being the base handle. It is important to find the correct size, feel, and weight to the pieces of the set to provide the best experience. This can probably only be done best by printing multiple different trials and deciding which is best based on experimental results.

Thirdly, with more time the box can be laser cut and etched from the designs I have already drawn up. Having a container is a practical and connotative aspect. It is extremely important to tie the set together as much as possible. For the size of the items I have created it is important to have them kept together in an orderly aesthetically pleasing manner. The top of the box hints at the design of the set as a whole with a simple silhouette of the original design. I hope to implement most of these plans, cost willing.